

THE STRUCTURE OF MATHEMATICAL REALITY

Phase I — Chapter 1

Mathematics Is Not About Numbers

Note Template

Self-Study Curriculum

How to use this template

Complete these notes **after** the reading and the lecture. The goal is not to copy phrases from the text — it is to reconstruct the ideas in your own language. Leave every item blank on first pass; fill it in from memory, then check.

Fill this in after reading both the lecture and the parallel reading. Write in your own words, transcription is not learning.

1. My previous working definition of mathematics was

write it out, even if it feels naive, you need the contrast later

2. The definition I'm now working with

your words, not the lecture's

3. The organizing principle behind the major fields

Each field is defined by what kind of _____ it _____.

- Algebra preserves:
- Analysis preserves:
- Geometry preserves:
- Topology preserves:
- Logic studies:

4. The idea that surprised me most

one sentence

5. The idea I'm not yet convinced by

one sentence, skepticism is a tool, not a failure

6. An example from my own work where I was extracting structure

without calling it that:
describe it concretely

7. Open question I'm carrying forward

what do you still want answered?
